A Robust UAV Hyperspectral Image Stitching Method Based on Deep Feature Matching

Yan Mo, Xudong Kang, *Senior Member, IEEE*, Puhong Duan, *Student Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Unmanned aerial vehicle (UAV) hyperspectral imaging has been extensively applied in various fields. However, due to the limited imaging width, hyperspectral images (HSIs) captured by UAV need to be stitched, so as to effectively cover the study area. In this article, an effective seamless stitching method with deep feature matching and elastic warp is proposed for HSIs, which consists of the following major steps. First, for each input HSI, a single-band gray-scale image is obtained by fusing the bands corresponding to the red, green, and blue wavelengths. Second, the feature points of each HSI are obtained with a robust VGG-style network and matched with a graph neural network. After point pairs are obtained, the next step is to estimate the transformation matrix of adjacent images, and a spectral correction method based on intrinsic decomposition is proposed to ensure the spectral consistency of adjacent images. In the final stage, a seam-cutting and multiscale blending strategy is adopted to ensure the spatial consistency of the stitching results. Experimental results on real HSIs show that the proposed method is superior to six representative image stitching approaches.**

***Index Terms*— Hyperspectral images (HSIs), image matching, image stitching, seam cutting, spectral correction, unmanned aerial vehicle (UAV).**

I. Introduction

HYPERSPECTRAL image (HSI) is a multiband image with a large number of bands that record the reflection characteristics of objects under different wavelengths [1], [2]. As a result, each type of object in HSIs has its unique spectral reflection curve. By taking this advantage, HSIs have been successfully applied to various applications, such as target detection [3], [4] and land-cover mapping [5], [6]. Hyperspectral remote sensing images are typically obtained by satellites equipped with hyperspectral imagers. However, the long revisit period and high altitude make it difficult for satellites to obtain high-quality multitemporal hyperspectral data, which severely limits its applications.

With the development of spectral imaging techniques, the system structure of the imaging spectrometers has become lighter and cheaper. Unmanned aerial vehicles (UAVs) equipped with imaging spectrometers have been widely used in remote sensing applications [7]–[9]. Compared with satellite remote sensing, UAVs can acquire HSIs with higher spatial resolution [10], [11]. However, the ground coverage area of each data acquisition is limited due to the fixed view angle of the imaging spectrometer and the low flight altitude of the UAV. In order to solve this problem, multiple HSIs can be stitched together, so as to effectively cover the study area in many remote sensing applications (e.g., geographical mapping, environmental monitoring, and disaster monitoring) [12].

Image stitching is the process of combining two or more images based on their overlapping regions to create a large and seamless image with wider viewing angle [13]. In general, image stitching consists of three major operations: image registration, radiometric normalization, and image blending [12]. Image registration is used to ensure that the images to be stitched are geometrically consistent [14]. Since the images to be stitched differ significantly in terms of radiation in most cases, the radiometric normalization plays an important role to ensure the illumination consistency of the stitching result [15]. The objective of image blending is to obtain a smooth transition from one image to another adjacent image [12]. In order to solve the problems faced in the major steps mentioned above, many algorithms have been developed. For example, geographic information can be used as a preprocessing step to obtain a rough registration result. Then, the rough registration result can be optimized by feature-based matching methods [16]. In [17], a global optimization combines an initial solution of the histogram extreme-point matching strategy that is applied to realize intensity consistency. Tian *et al.* [18] propose a six-parameter diagonal model to compensate for the radiometric differences by spectral transformation between adjacent images.

Different from general image stitching methods, HSIs' stitching considers not only the alignment of spatial information, such as edges and textures in each band, but also the spectral continuity of the stitched HSI. In the spatial registration step, most of the HSIs captured by satellites are registered by using geographic information. However, for UAVs without the ground real-time kinematic (RTK) system, the positional accuracy of GPS may decrease. In this situation, the geographic information-based registration methods cannot

Manuscript received September 1, 2021; revised October 21, 2021; accepted October 24, 2021. Date of publication October 28, 2021; date of current version February 14, 2022. This work was supported in part by the Major Program of the National Natural Science Foundation of China under Grant 61890962; in part by the National Natural Science Foundation of China under Grant 61601179, Grant 61871179, and Grant 61801178; in part by the Natural Science Foundation of Hunan Province under Grant 2019JJ50036; in part by the Scientific Research Project of the Hunan Education Department under Grant 19B105; in part by the Innovative Provincial Construction Project of Hunan Province under Grant 2020GK2038; in part by the Fund of Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province under Grant 2018TP1013; and in part by the Fund of Science and Technology Program of Guangdong Province under Grant 2018B010107001. *(Corresponding author: Xudong Kang.)*

The authors are with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha 410082, China (e-mail: moyan@hnu.edu.cn; xudong_kang@163.com; puhong_duan@hnu.edu.cn; shutao_li@hnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3123980

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

obtain a satisfactory performance [19]. In the radiometric normalization step, also known as the spectral correction step for HSIs, semiempirical correction methods are widely researched [20]. These methods generally consist of mathematical simplifications of the important radiometric effects, such as the models of the atmosphere. However, the inversion of parameters in these methods is usually complex.

In recent years, several representative UAV-based HSIs' stitching methods have been proposed. Fang *et al.* [21] design a UAV-based hyperspectral imaging system, which integrated a stitching method based on sparse bundle adjustment (SSBA). In [22], a two-step registration strategy is adopted. The UAV-based HSIs are preregistered using geographic information, and then, feature extraction and matching methods are adopted to optimize the registration results. To eliminate mismatching of feature points and improve the quality of stitching results, Zhang *et al.* [23] introduce a feature matching algorithm named multiscale top K rank preservation (mTopKRP) for HSIs' stitching. In general, all these methods have not considered the spatial and spectral consistencies of stitching results simultaneously.

To overcome this problem, a novel HSIs' stitching method is presented in this article, which consists of the following steps. First, a band fusion method is performed on each HSI to obtain a single-band gray-scale image. Then, the fused gray-scale image pairs are first fed into the SuperPoint network to extract the feature points and then fed into the SuperGlue network to obtain the matching relationship of feature points. This step makes an important role in ensuring the spatial consistency of the stitching result. Finally, intrinsic decomposition-based spectral correction and a seam-cutting-based multiscale blending strategy are employed to ensure spectral consistency. Experimental results on real UAV hyperspectral data demonstrate that the proposed method can obtain superior performance with respect to other state-of-the-art stitching approaches in terms of objective and subjective evaluations. The main innovations of this work can be concluded as follows.

1) A novel deep learning-based feature extraction and matching framework is developed for the feature extraction and matching of adjacent HSIs, which makes an important role in ensuring the spatial consistency of the stitching results.
2) An intrinsic decomposition-based spectral correction method and a seam-cutting blending method are proposed to ensure the spectral consistency of the stitching result.
3) A novel hyperspectral dataset that consists of 37 hyperspectral trips is developed in this work, which can be very helpful for other researchers in this field.

The remainder of this article is organized as follows. Section II briefly reviews some researches for the critical steps of image stitching. In Section III, the proposed HSIs' stitching method is introduced in detail. Sections IV and V are the experimental results and conclusions, respectively.

II. Related Work

In this section, we briefly review the major steps of HSIs' stitching, i.e., image registration, spectral correction, and image blending.

A. Image Registration

The process of aligning two or more images in the same scene captured from different viewpoints, times, or sensors is regarded as image registration [24]. The traditional image registration methods can be divided into area-based methods and feature-based methods. The area-based methods aim to realize image registration by using the intensity information of entire images [25]. The feature-based methods typically employ a three-step process, i.e., feature detection, feature description, and feature matching [24]. These traditional image registration methods have been demonstrated to be both efficient and robust. As a result, they have been widely used in a variety of fields [26]–[28]. However, the design of traditional image registration frameworks requires prior knowledge as guidance information.

In recent years, deep learning also has been widely applied in image registration. For example, learned invariant feature transform (LIFT) implements feature point detection, direction estimation, and feature description tasks using an effective end-to-end style CNN network [29]. LF-Net designs a novel two-branch configuration and iteratively creates a virtual target response that allows training without the handcraft priorities from scratch [30]. DeTone *et al.* [31] propose a self-supervised framework called SuperPoint for training interest point detectors and descriptors for multiple view geometrical problems. Ofir *et al.* [32] propose a new method for matching feature points from different spectral channels based on their deep descriptors. For a detailed and comprehensive review on deep learning-based image registration, please refer to [33].

B. Spectral Correction

Due to the bidirectional reflectance distribution function (BRDF) effect [34], the gray values of different UAV HSIs in the same imaging area will be quite different. Stitching source HSIs directly without spectral correction will lead to spectral inconsistency of the stitching results, which has a negative impact on subsequent applications.

The spectral correction methods of HSIs can be classified into two categories: one is the semiempirical methods; the other is the statistical methods. The semiempirical methods describe the reflection mechanism between light and the surface of the objects. Such methods are often used for spectral correction of HSIs acquired by satellites or fixed-wing aircraft. However, the inversion of parameters in the semiempirical methods is complex and requires some prior information as guidance [35], [36]. The other is the statistical methods, which are usually based on image intensity distribution [37]–[39]. These methods are simple and efficient and, thus, are more suitable for UAV-based HSIs' stitching.

C. Image Blending

Humans are extremely sensitive to gradient changes of image gray values. As a result, the stitching seams will significantly reduce the quality of the stitching result [40]. Image blending can effectively reduce gradient differences, eliminate artifacts caused by inconsistent information in overlapping areas, and improve the overall quality of the stitched image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

Based on their principles, image blending methods can be divided into three categories: pixel-weight methods, seam-cutting-based methods, and transform domain-based methods.

1) Pixel-Weight Methods: For the pixel-weight methods, a weighted average of the inputs is used to obtain each output pixel, which can be expressed as follows:

$$I_b = \omega I_1 + (1-\omega) I_2 \tag{1}$$

where I_1 and I_2 are images to be blend and $0 \leq w \leq 1$ is the weight map of I_1. A variety of methods have been proposed to estimate the optimal weights, most of which use distance as the weighting standard. In other words, the closer the distance, the greater the weight. For example, Fisher *et al.* [41] design a classical linear weighting model named inverse distance weighting (IDW). In [42], a nonlinear weight calculation algorithm based on cosine distance is proposed to solve the problem of the poor mixing effect of buffer boundary in a linear model.

2) Transform Domain-Based Methods: In addition to the image domain, the weighting average can be performed in the transform domain, such as the gradient and Fourier domain. For example, in [43], the gradient-based method blends images using a weighted combination according to a normalized lookup table. To reduce artifacts in the stitching result, Patidar and Jain [44] and Su *et al.* [45] apply the pixel averaging method in the Fourier domain. In [46], a flexible image-blending method is proposed that defined a novel weighting function of the geometry, temporal order, and user control inputs.

3) Seam-Cutting-Based Methods: The seam-cutting-based methods can find an optimal seam that minimizes the difference near the overlapping areas, so as to eliminate parallax. Assume that O represents the overlapping region of two adjacent images; a seam is defined as the connection path through the overlapping region, starting with the first row and ending with the last. By defining a specific energy function, each pixel in the overlapping region is given an energy value. As a result, the problem of finding the optimal seam equals to the problem of minimizing the total energy of the seam as follows:

$$s^* = \min_s \text{Cost}(s) = \min_s \sum_{i=1}^{N_s} c(I(s_i)) \tag{2}$$

where $I(s_i)$ is the ith pixel on seam s and N_s denotes the number of pixels in s. In a variety of applications, seam-cutting-based methods have achieved competitive results [47]–[50].

Generally, most of the existing image blending methods have been widely used in image stitching since they can eliminate part of the seams caused by image edges. Among these methods, seam-cutting-based methods tend to obtain better performances since they can effectively avoid information conflict in overlapping regions.

III. Proposed Method

Fig. 1 presents the main steps of the proposed method. First, a fused gray-scale image is generated by band fusion.

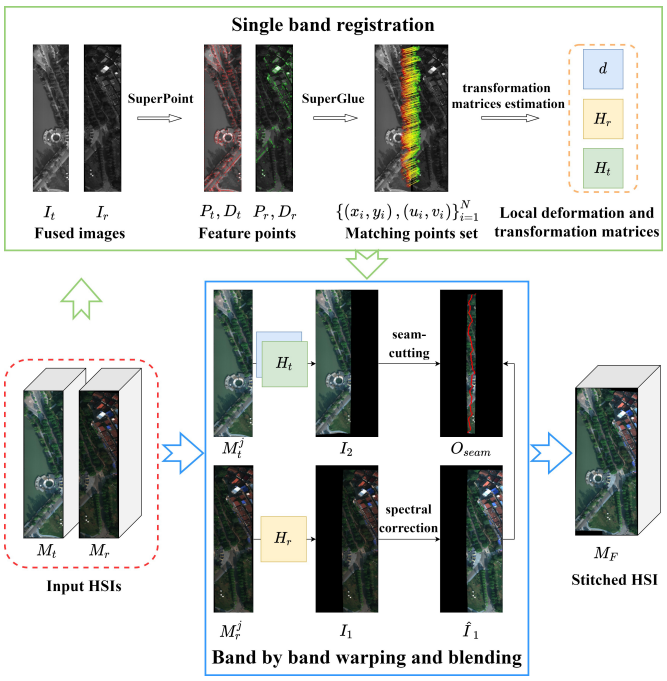


Fig. 1. Schematic figure of the proposed stitching method.

Then, the fused gray-scale image pairs are first fed into the SuperPoint network to extract feature points and then fed into the SuperGlue network to obtain matching point pairs. After that, the transformation matrix is estimated by robust elastic image warping. According to the estimated transformation matrix, the input HSIs are warped into the coordinate system of the stitching result. Next, based on the principle of intrinsic decomposition, band-by-band spectral correction of warped HSIs is implemented. Finally, the seam-cutting and multiscale blending strategies are used to obtain the final optimized stitching result.

A. Band Fusion

Because of the high spectral dimension of HSIs, using the original HSIs for feature points' extraction and matching will result in a massive computational cost. Furthermore, as shown in Fig. 2, there is a significant difference in image features among different HSI bands. As a result, for the extraction and matching of feature points, a single-band fused image is obtained to reduce the computing cost and improve the performance of spatial registration.

Specifically, given an input HSI M, the corresponding single-band fused gray-scale image I can be expressed as follows:

$$I = \frac{0.3}{n_r} * \sum_{i=r_1}^{r_2} M^i + \frac{0.59}{n_g} * \sum_{j=g_1}^{g_2} M^j + \frac{0.11}{n_b} * \sum_{k=b_1}^{b_2} M^k \tag{3}$$

where $[r_1, r_2]$, $[g_1, g_2]$, and $[b_1, b_2]$ represent the bands corresponding to the red, green, and blue wavelengths, respectively. n denotes the number of bands in different wavelength ranges.

B. Extraction and Matching of Feature Points

To obtain robust feature points and high-accurate matching point pairs, two different deep networks are adopted in our work, i.e., the SuperPoint network and the SuperGlue network.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

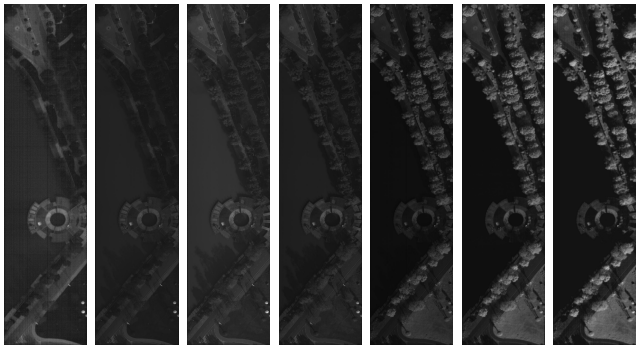


Fig. 2. Single-band images corresponding to different wavelengths of "Waterfront Square" data. (a) 400 nm. (b) 500 nm. (c) 600 nm. (d) 700 nm. (e) 800 nm. (f) 900 nm. (g) 1000 nm.

SuperPoint is constructed by a shared VGG-style encoder network and two different decoders, i.e., feature points' decoder and descriptors' decoder [31]. The dimension of the input image is reduced by the encoder. Following the encoder, the feature points' detection and the feature description tasks are simultaneously processed by two different decoders. This network structure enables shared computing and representation between two tasks. During the training phase, the SuperPoint network uses a self-supervised strategy. It is pretrained in generated graphs (simple shapes, such as lines, squares, and checkerboards) with labeled interest points and then uses a novel homographic adaptation procedure to automatically label images from the unlabeled domain. Besides, the homomorphic adaptation strategy is used in the training process to improve repeatability under viewpoint change. Thus, the network has reliable feature point extraction and representation capabilities in real images.

Traditional feature point matching methods usually consider only the position and descriptor of feature points. However, long-term features' aggregation within and across images, such as the spatial and visual relationship between feature points (self-similar feature points, statistical co-occurrence feature points, or adjacent feature points), are also important for feature points matching. The SuperGlue network designs an attentional graph neural module to optimize initial local feature descriptors by letting these features communicate with each other. Then, the optimal matching module generates a match confidence matrix to obtain the final optimal matching point pairs. The combination of two modules enables the SuperGlue network to eliminate ambiguity and achieve a competitive matching result [51].

As shown in Fig. 3, given two overlapped images, reference image I_r and target image I_t, the corresponding feature points P_r and P_t and their descriptors D_r and D_t can obtained by using two pretrained SuperPoint networks. Then, all feature points P and descriptors D are fed into a Superglue network to estimate their correlation and obtain a score matrix. Finally, the optimal match of each point $\{(x_i, y_i), (u_i, v_i)\}_{i=1}^{N}$ can be obtained by selecting the maximum value of each row in the score matrix, where (x_i, y_i) and (u_i, v_i) are the pixel coordinates and N is the total number of matching point pairs.

C. Robust Elastic Image Warping

After matching point pairs are obtained, the subsequent processing will estimate the geometric transformation matrix between input images and align them to a common coordinate system. Traditional image warping approaches calculate a global transformation matrix for each input image. For ideal cases when the camera translation is negligible, these methods work well [52]. However, undesirable artifacts may appear in the warping results for general cases with parallax. To improve the quality of warped images, some approaches propose a local warping field according to the point pairs between images [53], [54]. However, the local warping field may cause projective distortions in the nonoverlapped regions. Besides, the calculation of these methods usually is a time-consuming task.

In order to realize high-speed, high-quality warping and avoid projection distortion, a robust elastic warping method that combines local elastic warp with global similarity transformation is adopted in this work. First, the global transformation matrix H can be expressed as follows:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{4}$$

where $\sim$ denotes equality up to a scale and $[\,u\ v\ 1\,]^T$ and $[\,x\ y\ 1\,]^T$ represent the coordinates of matching point pairs. This formula can be solved by using direct linear transformation (DLT) [55] or the method presented by AutoStitch [52].

Then, the local deformation can be denoted as $d(x, y) = (g(x, y), h(x, y))^T$, where g and h represent horizontal and vertical deformations, respectively. The thin plate spline [56] with a radial basis function [57] is adopted to formulate the image deformation since it is a powerful tool in elastic warping and can achieve good performance. The deformation can be expressed as follows:

$$d(x, y) = \sum_{i=1}^{n} \omega_i \phi_i(x) + \alpha_1 x + \alpha_2 y + \alpha_3 \tag{5}$$

where $\phi_i(x)$ is the radial basis function, and $w = (\omega_1, \ldots, \omega_n)^T$ and $a = (\alpha_1, \alpha_2, \alpha_3)^T$ are the coefficients of $\phi_i(x)$, which can be obtained by solving a linear equation system.

However, extrapolating the warps to nonoverlapping regions may suffer from the overfitting problem, which refers to projection distortion. As a result, by gradually decreasing the deformation function $d(x, y)$ to 0 across the nonoverlapping regions, the local warp can smoothly transition to the global transformation H. The suppressed deformation $d_s(x, y)$ is represented as follows:

$$d_s(x, y) = \eta d(x, y) \tag{6}$$

where η is a scale parameter; it will change from 1 to 0 when (x, y) approaches the nonoverlapping areas gradually. The expression of η is given as follows:

$$\eta = 1 - \frac{\max(0, x - x_u, x_l - x, y - y_u, y_l - y)}{s_d} \tag{7}$$

where $[x_l, x_u]$ and $[y_l, y_u]$ are the boundaries of the overlapping area in the horizontal and vertical directions. s_d is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

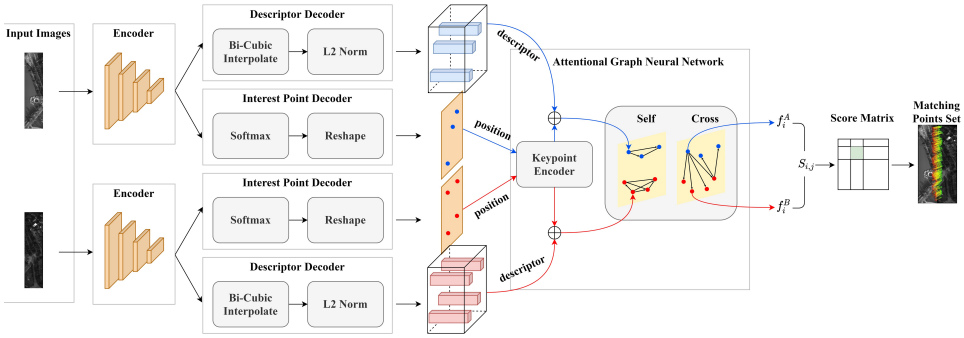


Fig. 3. Feature points' extraction and matching architecture.

pixel width of the smooth transition, which is set to be directly proportional to the maximum bias.

To speed up the computation, a uniform grid mesh of $B_x \times B_y$ blocks is introduced over the image plane. The deformations are calculated on the mesh nodes first and then linearly interpolated to other pixels. Using a deformation $d_s(x, y) = (g_s(x, y), h_s(x, y))^T$, I_t can be warped to $I_2^{\sim}$ to align with I_r. Given a point (x_0, y_0) in I_t and the corresponding point (x, y) in $I_2^{\sim}$, the local elastic warp can be expressed as follows:

$$\begin{cases} x_0 = x - g_s(x, y) \\ y_0 = y - h_s(x, y). \end{cases} \tag{8}$$

Since the similarity transformation is effective at mitigating projective distortions, combining local elastic warp with global similarity transformation can create an accurate alignments stitching field without projective distortions. The similarity transformation S is calculated by minimizing the following function:

$$J_A = \sum_{i=1}^{n_r} \left\| h_s P^i_{(u,v)} - P^i_{(x,y)} \right\|^2 \tag{9}$$

where $h_s = \begin{pmatrix} c & -s & t_x \\ s & c & t_y \end{pmatrix}$, $\left\{ P^i_{(u,v)}, P^i_{(x,y)} \right\}_{i=1}^{N}$ are the matching point pairs, and the global transformation of target image is represented as follows:

$$H_t = \mu_h H + \mu_s H_s \tag{10}$$

where

$$H_s = \begin{bmatrix} c & -s & t_x \\ s & c & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

where $\mu_h, \mu_s \in (0, 1)$ are the weighting coefficients with the constraint $\mu_h + \mu_s = 1$. The transformation of reference image I_r is calculated as follows:

$$H_r = H_t H^{-1}. \tag{11}$$

D. Intrinsic Decomposition-Based Spectral Correction

Due to various interferences, such as the change of atmosphere illuminations, the illumination conditions of the input HSIs are different, which may result in inconsistent gray values and edge artifacts of the stitching results. Hence, the spectral correction of the warped images should be considered. In other words, the spectra of the same objects in stitched HSI should be consistent. Instead of solving a complex semiempirical model, this article proposes a simple and effective spectral correction method to ensure the spectrum consistency of the stitching result. Specifically, the HSI M can be decomposed into the material-dependent property R and the light-dependent property L as follows:

$$M^n = R^n \times L \tag{12}$$

where n represents the nth band of the HSI. According to the overlap region mask O, the correction parameter κ for each band can be calculated as follows:

$$\kappa^n = \frac{\sum_{i=1}^{N_P} \left(I_1^i \cdot O \right)}{\sum_{i=1}^{N_P} \left(I_2^i \cdot O \right)}, \quad n = 1, 2, \ldots, C \tag{13}$$

where I is the nth band gray-scale image of HSI M and C denotes the total number of channels in the M. N_P represents the number of pixels in O. The warped target image I_2 is corrected as $\hat{I}_2 = \kappa I_2$.

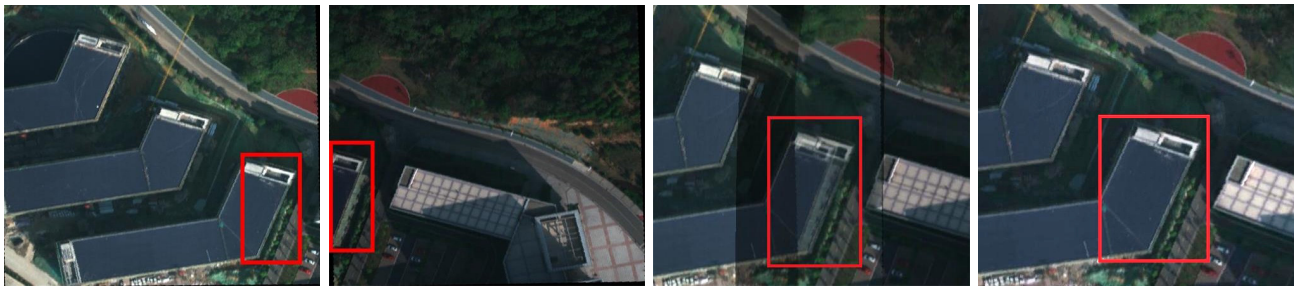
E. Seam-Cutting and Multiscale Blending

As shown in Fig. 4(a) and (b), since the flight altitude of rotor UAV is usually lower than 500 m, parallax may appear between adjacent HSIs, i.e., spatial information inconsistency in the overlapping areas. After image warping, the target image has been aligned with the reference image as accurately as possible. However, as shown in Fig. 4(c), ghosting artifacts still exist in the stitching result.

The seam-cutting strategy for image stitching is first proposed by Gao *et al.* [58] and has been proven to be an effective method for parallax image stitching. Therefore, in this work, the seam-cutting-based strategy is adopted to address the ghosting artifacts caused by parallax. After image warping and spectral correction, reference image I_1, corrected target image $\hat{I}_2$, and their overlapping area map O are obtained. A seam-cutting map O_{seam} can be obtained by labeling the points in O. Specifically, assuming point $p \in O$, assign p a label $l_p \in \{0, 1\}$, and $l_p = 0$ indicates that this point is designated as the target image. Conversely, $l_p = 1$ indicates that this point is designated as the reference image. The above calculations can be expressed as a process that minimizes the following energy functions:

$$E(l) = \sum_{p \in O} D_p(l_p) + \sum_{(p,q) \in N} S_{p,q}(l_p, l_q) \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 4. Blending comparison. (a) and (b) Input images with parallax. (c) Stitching result without seam-cutting blending. (d) Stitching result of our method.

where D and S represent data term and smoothness term, respectively. p and q are adjacent pixels, defined over a four-connected neighborhood. D is defined as follows:

$$\begin{cases} D_p(1) = 0, D_p(0) = \mu, & \text{if } p \in \partial I_1 \cap \partial O \\ D_p(0) = 0, D_p(1) = \mu, & \text{if } p \in \partial I_2 \cap \partial O \\ D_p(0) = D_p(1) = 0, & \text{otherwise} \end{cases} \tag{15}$$

which represents the cost of assigning a label to a pixel, where μ is a very large penalty to avoid mislabeling, $\partial I_k \cap \partial O$ is the common boundary between overlapping area O and image $I_k (k = 1, 2)$, and the data term $D_p(l_p)$ fixes the endpoints of the seam as the intersections of the two common boundaries. S is defined as

$$S_{p,q}(l_p, l_q) = \frac{1}{2}|l_p - l_q|(I_d(p) + I_d(q)) \tag{16}$$

which denotes the cost of assigning a pair of labels to a pair of pixels. I_d means the Euclidean-metric difference map

$$I_d(\cdot) = \left\| I_1(\cdot) - \hat{I}_2(\cdot) \right\|_2. \tag{17}$$

The energy function (14) is minimized by the graph cut algorithm [40]. More details about seam cutting can be found in [47] and [58].

To improve the visual effect of stitching results, the Laplacian pyramid with weights is utilized to blend images. First, Laplacian pyramids are constructed for reference and target images, respectively. Then, the Gaussian pyramid is constructed for the seam-cutting weight map O_{seam}. When the image pyramids are obtained, the final stitching result M_F is calculated as follows:

$$M_F = \texttt{recon}\left(\sum_{n=1}^{N_l} \left(1 - G^n_{\text{seam}}\right) L^n_1 + G^n_{\text{seam}} L^n_2 \right) \tag{18}$$

where $\texttt{recon}(\cdot)$ refers to the pyramid reconstruction operation. G_{seam} represents the Gaussian pyramid of the seam-cutting map, L_1 and L_2 represent the Laplacian pyramids of I_1 and $\hat{I}_2$, respectively. $N_l = [((\log(\min(m, n)))/(\log(2)))]$ is the total number of pyramid layers, where $m \times n$ is the spatial size of input images.

After single-band stitching is completed, the input HSIs are warped, corrected, and blended band by band to get the final stitched HSI, as shown in Fig. 1. To better illustrate the stitching process, Algorithm 1 shows an algorithm flowchart. It should be mentioned that, for the proposed method, multiple HSIs can be stitched together by stitching adjacent strips successively. The proposed method can be also extended for the stitching of more than two images simultaneously. In order to fuse more than two images simultaneously, two major steps of the proposed method should be modified. One is the step of extracting and matching feature points, and the other is the step of image warping. Specifically, the feature points of an image should be matched with the feature points extracted from all other images. Furthermore, in the matching and wrapping steps, all feature matches must be handled as a whole. However, this will dramatically increase the demand for computing memory and, thus, not adopted in this work.

Algorithm 1 Two HSIs' Stitching

Input: Hyperspectral reference image M_r and hyperspectral target image M_t with C channels

1: Band fusion: calculate representative images I_r and I_t based on Eq. (3)
2: Extract feature points P_r, P_t and corresponding feature descriptor D_r, D_t by SuperPoint
3: Obtain a matching points set $\{(x_i, y_i), (u_i, v_i)\}_{i=1}^{N}$ using SuperGlue
4: Using Algorithm 2 to calculate the local deformation d and transformation matrices H_t, H_r
5: **for** j=1 to C **do**
6: Initialization: input single band images M_r^j, M_t^j
7: Obtain the warped reference image I_1 using H_r
8: Obtain the local warped target image $I_2{}^{\sim}$ using d
9: Transfer $I_2{}^{\sim}$ to I_2 by using H_t
10: Get the overlapping area O
11: Warped target image spectral correction: $\hat{I}_2 = \kappa I_2$, κ is obtained by Eq. (13)
12: Seam-cutting map I_{seam} calculation by Eq. (14)
13: Image blending: using Eq. (18) to calculate single band stitching result M_F^j
14: $j = j + 1$
15: **end for**

Output: Stitched HSI M_F

IV. Experiments

In this section, the stitching capability of the proposed method is evaluated on real UAV HSIs. Several image stitching methods are adopted for comparison. All experiments are conducted on a desktop with 3.4-GHz AMD Ryzen 2600 CPU and 64-GB RAM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
KEY TECHNIQUE INDEXES OF NANO-HYPERSPEC

Parameter	Wavelength range	Spatial bands	Spectral bands	Dispersion/Pixel	FWHM slit image	f/#
Value	$400-1000nm$	640	270	$2.2nm/pixel$	$6nm$	2.5
Parameter	Entrance slit width	Camera technology	Detector pixel pitch	Focal length	FOV	IFOV
Value	$20\mu m$	CMOS	$7.4\mu m$	12nm	22°	0.61mrad

TABLE II
DATA INTRODUCTION

NO.	Dataset Name	Data Collection Location	Strip Numbers	Data Size	Scene Type	Major Types of Land Cover	Notes
1	Waterfront Square	Yueyang, Hunan Province, China	2	671×2645×270	City	Roads and vegetation	Large spectral differences
2	Qianmu Lake	Yueyang, Hunan Province, China	6	687×3428×270	City	Water, buildings and vegetation	Difficult to extract feature points Large spectral differences Irregular distortion and low overlap rate
3	College	Yueyang, Hunan Province, China	3	666×4014×270	City	Buildings, roads and vegetation	Long strip length and irregular shape Slight spectral difference
4	Riverbed	Dongying, Shandong Province, China	6	655×7738×270	Wetland	Bare soil	Difficult to extract feature points Slight spectral difference
5	Wetland	Dongying, Shandong Province, China	6	640×1250×270	Wetland	Wetland vegetation	Ambiguous feature points Slight spectral difference
6	Aquafarm	Changsha, Hunan Province, China	4	640×5004×270	Fishpond	Water and fallow land	Slight spectral difference
7	Farmland	Changsha, Hunan Province, China	2	643×5458×270	Farmland	Rice and water	Long strip length and irregular shape Slight spectral difference
8	Nursery Base	Changsha, Hunan Province, China	2	640×2240×270	Woodland	Woodland and water	Slight spectral difference
9	Dike	Changsha, Hunan Province, China	4	640×6492×270	Farmland	Fallow land, bare soil and water	Slight spectral difference
10	Experimental Plot	Changsha, Hunan Province, China	2	640×752×270	Farmland	Farmland and crops	Large spectral differences Heavy noise and geometric distortion

Algorithm 2 Robust Elastic Image Warping

Input: Matching points set $\{(x_i, y_i), (u_i, v_i)\}_{i=1}^{N}$, representative images I_r and I_t

1: Estimate the global transformation H between I_r and I_t using DLT [55]
2: Calculate the local deformation d using Eq. (6)
3: Estimate the similarity transformation H_s by Eq. (9)
4: $H_t = \mu_h H + \mu_s H_s$
5: $H_r = H_t H^{-1}$

Output: Local deformation d, transformation matrices H_t, H_r

(a) (b)

Fig. 5. (a) DJI M600pro equipped with a Headwall-nano hyperspectral imager. (b) Data collection in the Yellow River Estuary Wetland Reserve, Dongying, Shandong province.

Six state-of-the-art stitching methods are adopted for comparison, including AutoStitch [52], APAP [53], ANAP [54], ELA [59], Single-perspective [60], and mTopKRP [23]. AutoStitch and Single-perspective use global homography alignment, while the other methods use local homography alignment. mTopKRP is an HSI stitching method that uses the mTopKRP algorithm to eliminate the mismatching before warping and blending the HSIs band by band. It should be noted that all parameters of the compared methods are fixed and set according to the corresponding publications or the default values in the public codes provided by the authors.

A. Introduction of Experimental Data

In order to validate the effectiveness of the proposed stitching method, 37 HSI strips were collected by the UAV (DJI M600pro) equipped with a hyperspectral imager (Headwall-nano), as shown in Fig. 5. Besides, Table I lists the key parameters of the hyperspectral imager. Table II provides a brief overview of the HSIs obtained in different scenarios.

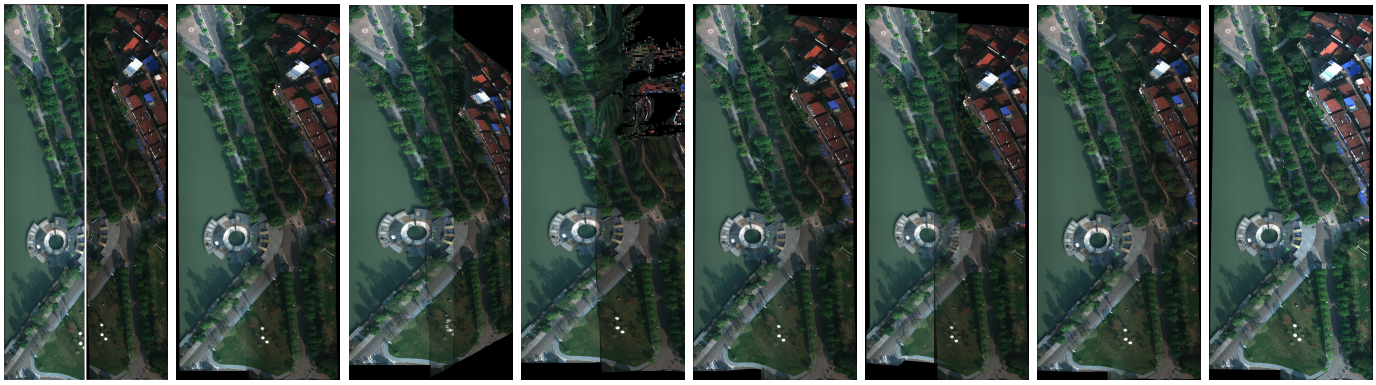
The "Waterfront Square," "Qianmu Lake," and "College" were collected in Qianmu Lake, Yueyang, Hunan province, China. They are representative urban scenes, with buildings, vegetation, water, and roads as the main ground objects. Because the sun elevation angle is small when collecting the data, the radiation differences among the two adjacent images are obvious. The same phenomenon exists in the "Qianmu Lake" and "College" data. The "Qianmu Lake" data contain a large amount of water, which will make feature extraction and matching more challenging. Besides, the UAV's flight path is slightly offset during data acquisition due to wind disturbance, resulting in a lower overlap rate and irregular distortion of the acquired HSIs. Compared with other scenes, the stitching of this data is relatively difficult.

The "Riverbed" and "Wetland" data were collected in Yellow River Estuary Wetland Reserve, Dongying, Shandong province, China. The two datasets represent two typical wetland scenarios: dry riverbed dominated by bare soil and a small amount of wetland vegetation, and swamp dominated by aquatic plants, such as water surface and reeds. The "Riverbed" data faces the same challenges as the "Qianmu Lake" data in terms of extracting feature points. In contrast, there is a problem that ambiguous feature points are found in the "Wetland" data due to a large number of similar feature points.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

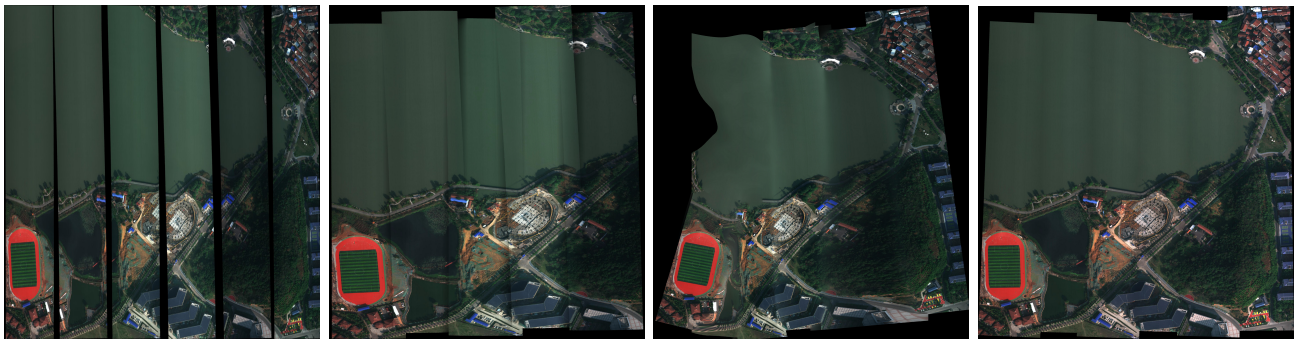
TABLE III
SUMMARY OF STITCHING RESULTS

NO.	Fig.	AutoStitch [52]	APAP [53]	ANAP [54]	ELA [59]	Single-perspective [60]	mTopKRP [23]	Ours
1	6	Obvious color differences	Serious distortion	Serious distortion	Obvious color differences	Unaligned	Obvious color differences	**Succeed**
2	7	Unfinished stitching Strip artifacts	Failed	Failed	Failed	Failed	Serious distortion	slight strip artifacts
3	8	Obvious local distortion	Failed	Failed	Local distortion	Unaligned	Local distortion	**Succeed**
4	9	Strip artifacts	Failed	Failed	Strip artifacts	Failed	Strip artifacts	**Succeed**
5	10	Slight artifacts	Serious distortion	Serious distortion	Slight color differences	Failed	Slight color differences	**Succeed**
6	11	Obvious local distortion Artifacts	Serious distortion	Serious distortion	**Succeed**	Artifacts	**Succeed**	**Succeed**
7	12	Failed	Failed	Failed	Strip artifacts	Failed	Strip artifacts	**Succeed**
8	13	Slight artifacts	**Succeed**	**Succeed**	**Succeed**	Failed	**Succeed**	**Succeed**
9	14	Obvious uneven brightness	Failed	Failed	Slight local distortion	Unaligned	**Succeed**	**Succeed**
10	15	Obvious artifacts Color differences	Obvious artifacts Color differences	Obvious artifacts Color differences	Obvious color differences Slight geometric distortion	Failed	Obvious color differences	**Succeed**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. "Waterfront Square" stitching results (pseudocolor images). (a) Source images. (b) AutoStitch [52]. (c) APAP [53]. (d) ANAP [54]. (e) ELA [59]. (f) Single-perspective [60]. (g) mTopKRP [23]. (h) Proposed method.



(a) (b) (c) (d)

Fig. 7. "Qianmu Lake" stitching results. (a) Source images. (b) AutoStitch [52]. (c) mTopKRP [23]. (d) Proposed method.

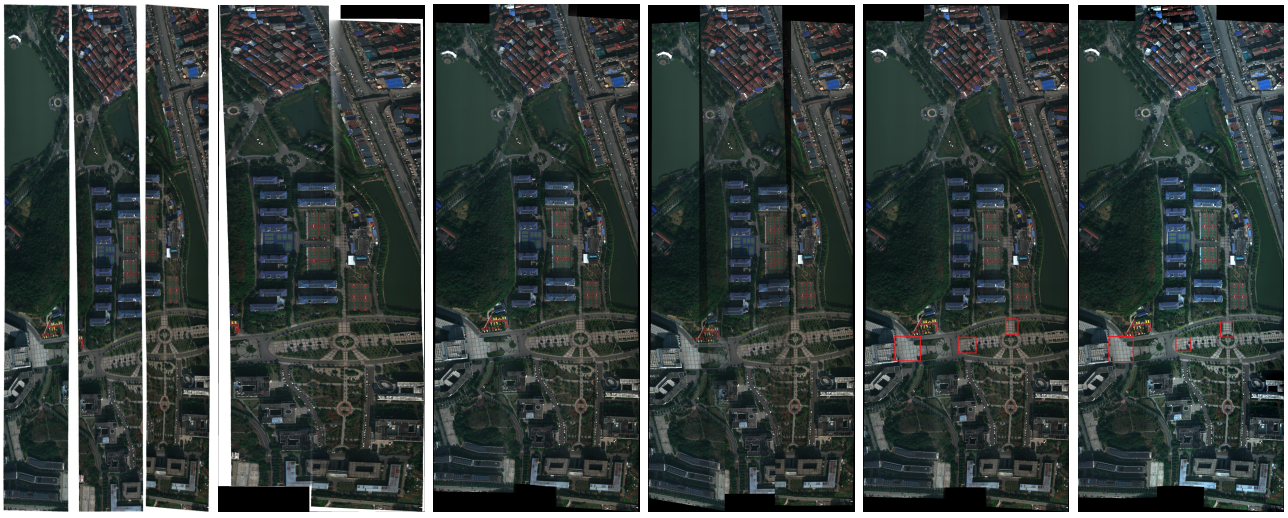
The remaining data were collected in Changsha, Hunan province. These datasets represent typical agricultural, fishery, and forestry scenarios that face the same challenges as the previous data. In addition, it should be noted that the "Experimental Plot" data have severe spectral distortions due to insufficient sunlight intensity.

These datasets contain a variety of typical land covers, and there are several critical issues in the process of UAV hyperspectral data collection, such as large spectral differences, difficult to extract feature points, heavy noise, and geometric distortion. Therefore, experiments on these datasets can effectively reflect the stitching performance and feasibility of different methods in real-world applications.

B. Spatial Consistency Analysis

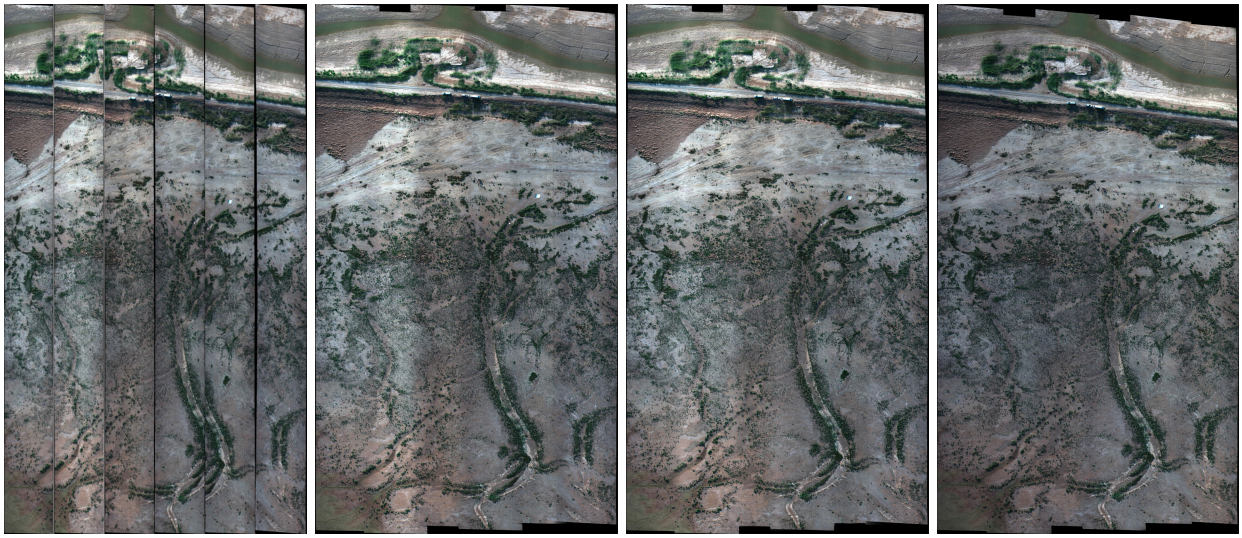
1) Visual Comparison: Figs. 6–15 present the pseudocolor images of all stitching results. Table III displays a summary of the stitching results; "Failed" in this table means that these methods failed in obtaining a stitching result, which could be presented for comparison. APAP, ANAP, and Single-perspective are failed in most cases because the image warping algorithm of APAP and ANAP is unsuitable for strip image stitching. When there is a lack of line features in the scene, the linear constraint stitching optimization strategy adopted by the Single-perspective method may fail. Therefore, the latter result analysis mainly focuses on the other three comparison methods: AutoStitch, ELA, and mTopKRP.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.



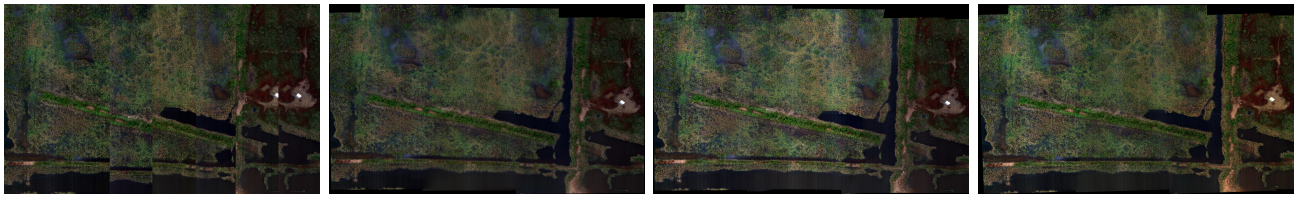
(a) (b) (c) (d) (e) (f)

Fig. 8. "College" stitching results. (a) Source images. (b) AutoStitch [52]. (c) ELA [59]. (d) Single-perspective [60]. (e) mTopKRP [23]. (f) Proposed method.



(a) (b) (c) (d)

Fig. 9. "Riverbed" stitching results. (a) Source images. (b) AutoStitch [52]. (c) mTopKRP [23]. (d) Proposed method.

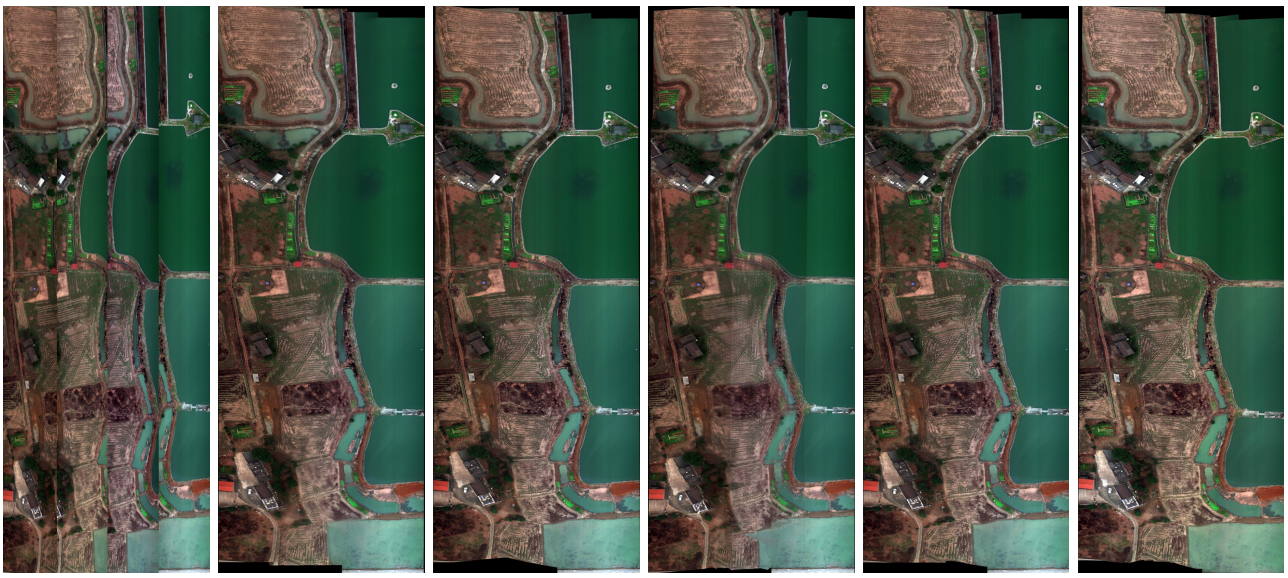


(a) (b) (c) (d)

Fig. 10. "Wetland" stitching results. (a) Source images. (b) AutoStitch [52]. (c) mTopKRP [23]. (d) Proposed method.

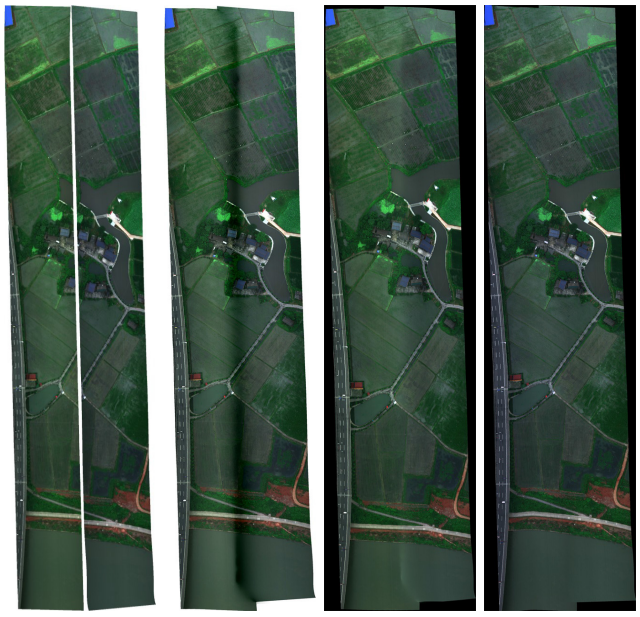
In urban scenes (NOs. 1–3), all these comparison methods have successfully stitched the "Waterfront Square" (NO. 1) data. However, there are significant spectral differences between the left and right parts of the stitching result. Since the irregular distortion and low overlap rate in "Qianmu Lake" (NO. 2) data, the AutoStitch method is unable to complete the stitching of all strip images. Besides, the stitching result of the AutoStitch method has obvious seams. Although the mTopKRP method completed the stitching on "Qianmu Lake" (NO. 2) data, the cost is a significant distortion of the stitching result. The stitching results of "College" (NO. 3) data are similar to "Waterfront Square" (NO. 1) data. The stitch results show spectral differences between the same types of ground object, as displayed in the red boxes in Fig. 8(e) and (f). All these comparison methods ensure spatial consistency of stitched results in wetland scenarios (NOs. 4 and 5). However, due to the large homogeneous areas in these data, the striping effect in the overlapping areas of the stitching results is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f)

Fig. 11. "Aquafarm" stitching results. (a) Source images. (b) AutoStitch [52]. (c) ELA [59]. (d) Single-perspective [60]. (e) mTopKRP [23]. (f) Proposed method.



(a) (b) (c) (d)

Fig. 12. "Farmland" stitching results. (a) Source images. (b) ELA [59]. (c) mTopKRP [23]. (d) Proposed method.

more obvious. In the remaining five data, "Farmland" (NO. 4) and "Dike" (NO. 5) data also have obvious strip artifacts. Furthermore, the spectral of "Experimental Plot" (NO. 10) is significantly different.

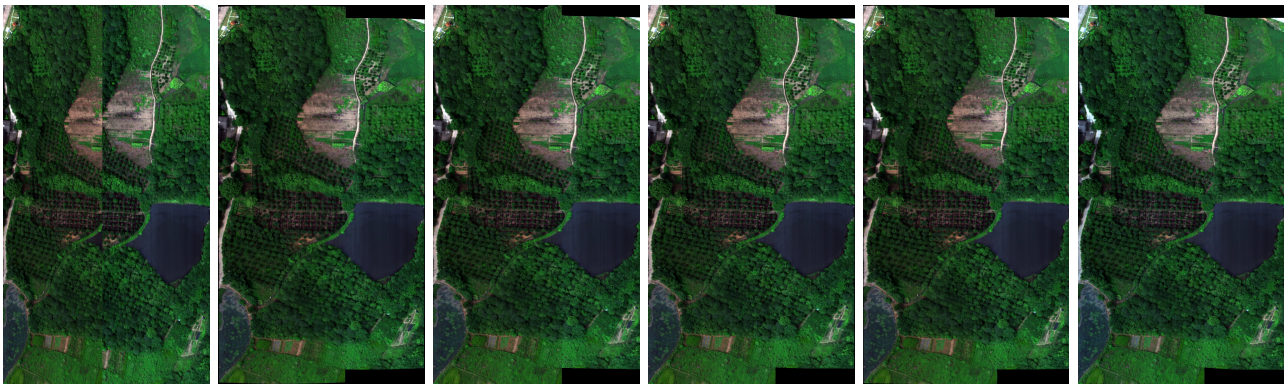
Conversely, the proposed stitching method successfully stitches all data. Moreover, the stitching results exhibit high spectral consistency, especially in "Waterfront Square," "College," and "Experimental Plot" data. As shown in Figs. 7, 10, 11, and 14, slight strip-like artifacts are observed on the water surface. The reason is that the proposed intrinsic decomposition-based spectral correction method performs not very well on homogeneous areas. In this situation, the small gray value differences of adjacent bands will be more obvious in these areas. Besides, the strip noise in the input HSIs cannot be removed by the proposed method and, thus, also may produce strip-like artifacts on the stitching results. In order to address this problem, one possible solution is performing image denoising on those homogeneous areas, so as to remove those artifacts.

2) Objective Comparison: The quantitative performance of different stitching methods is evaluated by the structural similarity index (SSIM) [61]. Specifically, after the images to be stitched are transformed into the same coordinate system, the SSIM index of the overlapping area is calculated to evaluate the spatial consistency of stitching results. If there are multiple overlapping areas, their average value is taken as the final SSIM value. ELA and mTopKRP, which have the better visual effect of stitching results, are selected as comparison methods. As can be seen from Table IV, the proposed method achieves the highest score in almost all data. Compared with previous methods, the proposed method has a significant improvement in terms of the SSIM index, particularly in HSIs with data NOs. 1, 2, and 4. In general, Table IV demonstrates that the proposed method performs better in ensuring the spatial consistency of the stitching results.

C. Spectral Consistency Analysis

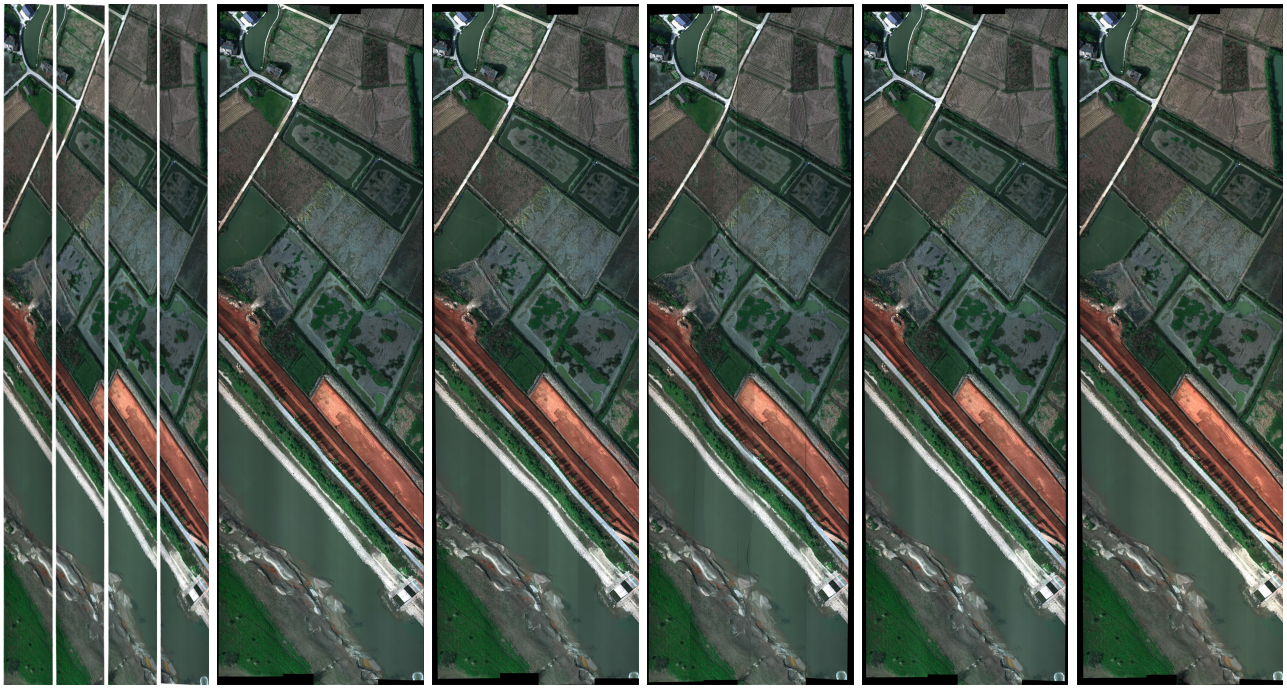
Spectral consistency is another important quality indicator for HSIs' stitching. Thus, four representative types of land cover, namely, roads, vegetation, water, and soil, are chosen from the "Waterfront Square" and "Lakeside" data. Their spectra curves are shown to evaluate the spectral consistency of the stitching results. Specifically, ten random sampling points are selected in the overlapping region of adjacent HSIs for each land cover type. The spectral analysis is performed on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.



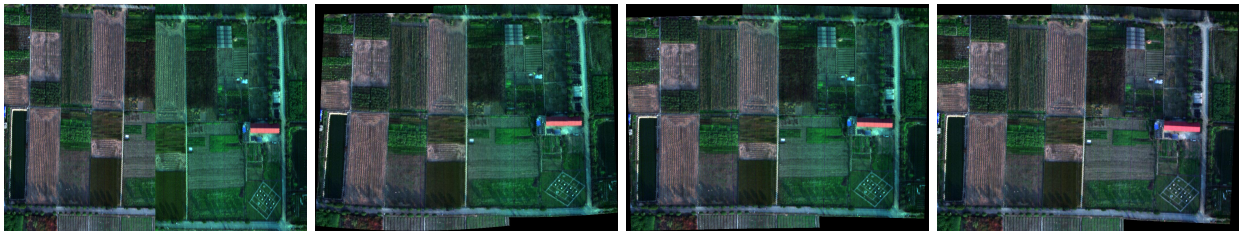
(a) (b) (c) (d) (e) (f)

Fig. 13. "Nursery Base" stitching results. (a) Source images. (b) AutoStitch [52]. (c) ELA [59]. (d) Single-perspective [60]. (e) mTopKRP [23]. (f) Proposed method.



(a) (b) (c) (d) (e) (f)

Fig. 14. "Dike" stitching results. (a) Source images. (b) AutoStitch [52]. (c) ELA [59]. (d) Single-perspective [60]. (e) mTopKRP [23]. (f) Proposed method.



(a) (b) (c) (d)

Fig. 15. "Experimental Plot" stitching results. (a) Source images. (b) AutoStitch [52]. (c) mTopKRP [23]. (d) Proposed method.

the 9×9 neighborhood centered on these points. As shown in Fig. 16, the spectral curves of the same type of land cover have large amplitude differences in adjacent uncorrected HSIs because these points are selected in the "Waterfront Square" data, which is affected by the sun's altitude angle during the collection process. However, it can be seen that the trend and amplitude values of the two spectral curves after correction with the proposed method are very similar in Fig. 16.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SSIM OF DIFFERENT STITCHING RESULTS

NO.	1	2	3	4	5	6	7	8	9	10
ELA [59]	0.9158	–	0.9046	0.8783	0.9582	0.9309	0.8857	**0.9951**	0.9242	0.9906
mTopKRP [23]	0.9154	0.8985	0.9169	0.8829	0.9797	0.9324	0.9407	0.9948	0.9494	0.9885
Ours	**0.9346**	**0.9169**	**0.9272**	**0.9357**	**0.9812**	**0.9383**	**0.9488**	0.9936	**0.9501**	**0.9912**

TABLE V
QUANTITATIVE ANALYSIS

	Roads		Vegetation		Water		Soil	
	Uncorrected	Corrected	Uncorrected	Corrected	Uncorrected	Corrected	Uncorrected	Corrected
RMSE	0.0532	**0.0082**	0.0745	**0.0112**	0.0083	**0.0048**	0.0227	**0.0156**
ED	0.6164	**0.0711**	0.8603	**0.0803**	0.0802	**0.0227**	0.1966	**0.0687**
SID	0.3451	**0.2771**	0.0507	**0.0473**	0.0309	**0.0296**	0.1147	**0.0925**
SAM	0.0450	**0.0391**	0.0623	**0.0522**	0.0241	**0.0133**	0.0370	**0.0249**

The other two types of land cover, water and bare soil, also achieve similar performances.

In addition, some quantitative analysis metrics are adopted to validate spectral consistency. The root mean square error (RMSE) and the Euclidean distance are two commonly used evaluation indexes, which can be used to measure the deviation between the observed value and the true values. Moreover, the spectral information divergence (SID) measures spectral similarity by comparing the likelihood distributions of the two spectra. It can be calculated as follows:

$$\mathrm{SID} = \sum_{i=1}^{C} p_i \log\left(\frac{p_i}{q_i}\right) + \sum_{i=1}^{C} q_i \log\left(\frac{q_i}{p_i}\right). \tag{19}$$

The spectral angle mapper (SAM) calculates the similarity between two arrays, and the result can be viewed as the cosine angle between the two arrays. The formula is given as follows:

$$\mathrm{SAM} = \cos^{-1}\left(\frac{\sum_{i=1}^{C} p_i q_i}{\sqrt{\sum_{i=1}^{C} p_i^2}\sqrt{\sum_{i=1}^{C} q_i^2}}\right) \tag{20}$$

where C is total number of bands for HSIs. $q_i \in (x, y)$ and $p_i \in (u, v)$ are the gray values for the reference spectrum and the test spectrum, respectively. The detailed experimental results are listed in Table V. Lower RMSE, ED, SID, and SAM values indicate that the stitching results have better spectral consistency. The proposed stitching method outperforms the uncorrected HSIs' stitching results in all indicators, demonstrating that it can ensure the spectral consistency of the stitching results.

D. Computational Efficiency

To evaluate the computational efficiency of different stitching methods, we calculated the running time of all methods on the "Waterfront Square" data. As shown in Table VI, it is clear that our method outperforms all the compared approaches in terms of computational efficiency.

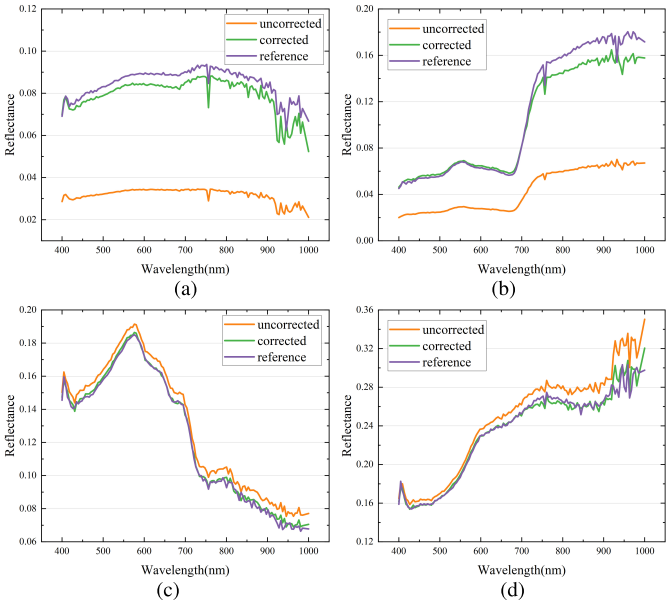


Fig. 16. Spectral curves of typical land covers. (a) Roads. (b) Vegetation. (c) Water. (d) Soil.

TABLE VI
RUNNING TIME COMPARISON (S)

AutoStitch	APAP	ANAP	ELA	Single-perspective	*m*TopKRP	Ours
100.90	103.52	112.41	110.37	102.92	112.55	**98.56**

According to the compared results, we can conclude that the proposed method can achieve efficient and accurate HSIs' stitching in various scenes, even in challenging cases with geometric distortion, ambiguous features, or heavy noise. However, the spectral correction of the proposed stitching method is based on the overlapping region; it may produce strip-like artifacts on the stitching results when the overlap rate is extremely low.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this work, a novel UAV HSI stitching method has been introduced. First, a band fusion method is adopted to obtain the single-band fused gray-scale images of input HSIs. Then, with the combination of SuperPoint and SuperGlue networks, the feature points are extracted and matched. After that, the transformation matrix is estimated by robust elastic image warping. Next, based on the principle of intrinsic decomposition, band-by-band spectral correction of warped HSIs is implemented. Finally, to merge the HSIs, a blending strategy combining seam-cutting and multiscale methods is used. A comprehensive evaluation of several real UAV HSIs validates that our method significantly outperforms other state-of-the-art stitching approaches. Furthermore, our method has reliable performance on a series of challenging cases.

References

[1] F. Luo, H. Huang, Z. Ma, and J. Liu, "Semisupervised sparse manifold discriminative analysis for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6197–6211, Oct. 2016.

[2] Y. Duan, H. Huang, and T. Wang, "Semisupervised feature extraction of hyperspectral image using nonlinear geodesic sparse hypergraphs," *IEEE Trans. Geosci. Remote Sens.*, early access, Sep. 15, 2021, doi: 10.1109/TGRS.2021.3110855.

[3] S. Yang, Z. Shi, and W. Tang, "Robust hyperspectral image target detection using an inequality constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3389–3404, Jun. 2015.

[4] F. Li, X. Zhang, L. Zhang, D. Jiang, and Y. Zhang, "Exploiting structured sparsity for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4050–4064, Jul. 2018.

[5] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[6] H. Sun, X. Zheng, and X. Lu, "A supervised segmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 2810–2825, 2021.

[7] P. Duan, P. Ghamisi, X. Kang, B. Rasti, S. Li, and R. Gloaguen, "Fusion of dual spatial information for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7726–7738, Sep. 2021.

[8] S. Zhang, X. Kang, P. Duan, B. Sun, and S. Li, "Polygon structure-guided hyperspectral image classification with single sample for strong geometric characteristics scenes," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 21, 2021, doi: 10.1109/TGRS.2021.3094582.

[9] P. Duan, J. Lai, J. Kang, X. Kang, P. Ghamisi, and S. Li, "Texture-aware total variation-based removal of sun glint in hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 359–372, Aug. 2020.

[10] X. Li *et al.*, "Detection and biomass estimation of phaeocystis globosa Blooms off southern China from UAV-based hyperspectral measurements," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 26, 2021, doi: 10.1109/TGRS.2021.3051466.

[11] X. Hu *et al.*, "SPNet: Spectral patching end-to-end classification network for UAV-borne hyperspectral imagery with high spatial and spectral resolutions," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 22, 2021, doi: 10.1109/TGRS.2021.3049292.

[12] X. Li, R. Feng, X. Guan, H. Shen, and L. Zhang, "Remote sensing image mosaicking: Achievements and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 4, pp. 8–22, Apr. 2019.

[13] R. Szeliski, "Image alignment and stitching: A tutorial," *Found. Trends Comput. Graph. Vis.*, vol. 2, no. 1, pp. 1–104, 2006.

[14] S. Dawn, V. Saxena, and B. Sharma, "Remote sensing image registration techniques: A survey," in *Proc. Int. Conf. Image Signal Process.* Berlin, Germany: Springer, 2010, pp. 103–112.

[15] A. Mills and G. Dudek, "Image stitching with dynamic elements," *Image Vis. Comput.*, vol. 27, no. 10, pp. 1593–1602, 2009.

[16] T. Suzuki, Y. Amano, and T. Hashizume, "Vision based localization of a small UAV for generating a large mosaic image," in *Proc. IEEE SICE Annu. Conf.*, Aug. 2010, pp. 2960–2964.

[17] R. Xie, M. Xia, J. Yao, and L. Li, "Guided color consistency optimization for image mosaicking," *ISPRS J. Photogram. Remote Sens.*, vol. 135, pp. 43–59, Jan. 2018.

[18] G. Yun Tian, D. Gledhill, and D. Taylor, "Comprehensive interest points based imaging mosaic," *Pattern Recognit. Lett.*, vol. 24, nos. 9–10, pp. 1171–1179, Jun. 2003.

[19] Y. Yuan, F. Fang, and G. Zhang, "Superpixel-based seamless image stitching for UAV images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1565–1576, Feb. 2021.

[20] S. Collings, P. Caccetta, N. Campbell, and X. Wu, "Techniques for BRDF correction of hyperspectral mosaics," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3733–3746, Oct. 2010.

[21] J. Fang, X. Wang, T. Zhu, X. Liu, X. Zhang, and D. Zhao, "A novel mosaic method for uav-based hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 9220–9223.

[22] H. Zhang, B. Zhang, Z. Wei, C. Wang, and Q. Huang, "Lightweight integrated solution for a UAV-borne hyperspectral imaging system," *Remote Sens.*, vol. 12, no. 4, p. 657, Feb. 2020.

[23] Y. Zhang, Z. Wan, X. Jiang, and X. Mei, "Automatic stitching for hyperspectral images using robust feature matching and elastic warp," *IEEE J. Sel. Topics Appl. Earth Observ., Remote Sens.*, vol. 13, pp. 3145–3154, 2020.

[24] B. Zitová and J. Flusser, "Image registration methods: A survey," *Image Vis. Comput.*, vol. 21, pp. 977–1000, Oct. 2003.

[25] X. Jiang, J. Ma, G. Xiao, Z. Shao, and X. Guo, "A review of multimodal image matching: Methods and applications," *Inf. Fusion*, vol. 73, pp. 22–71, Sep. 2021.

[26] F. Endres, J. Hess, N. Engelhard, J. Sturm, D. Cremers, and W. Burgard, "An evaluation of the RGB-D SLAM system," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2012, pp. 1691–1696.

[27] J. L. Schonberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 4104–4113.

[28] S. Paul and U. C. Pati, "Remote sensing optical image registration using modified uniform robust SIFT," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 9, pp. 1300–1304, Sep. 2016.

[29] K. M. Yi, E. Trulls, V. Lepetit, and P. Fua, "LIFT: Learned invariant feature transform," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 467–483.

[30] Y. Ono, E. Trulls, P. Fua, and K. M. Yi, "LF-Net: Learning local features from images," in *Proc. Intl. Conf. Neural Inf. Process. Syst.*, 2018, pp. 6237–6247.

[31] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 224–236.

[32] N. Ofir, S. Silberstein, H. Levi, D. Rozenbaum, Y. Keller, and S. D. Bar, "Deep multi-spectral registration using invariant descriptor learning," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 1238–1242.

[33] R. Feng, H. Shen, J. Bai, and X. Li, "Advances and opportunities in remote sensing image geometric registration: A systematic review of state-of-the-art approaches and future research directions," *IEEE Geosci. Remote Sens. Mag.*, early access, Jun. 30, 2021, doi: 10.1109/MGRS.2021.3081763.

[34] D. Schläpfer, R. Richter, and T. Feingersh, "Operational BRDF effects correction for wide-field-of-view optical scanners (BREFCOR)," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 1855–1864, Apr. 2015.

[35] J. Weyermann, A. Damm, M. Kneubuhler, and M. E. Schaepman, "Correction of reflectance anisotropy effects of vegetation on airborne spectroscopy data and derived products," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 616–627, Jan. 2014.

[36] F.-M. Bréon and E. Vermote, "Correction of MODIS surface reflectance time series for BRDF effects," *Remote Sens. Environ.*, vol. 125, pp. 1–9, Oct. 2012.

[37] S. J. Kim and M. Pollefeys, "Robust radiometric calibration and vignetting correction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 4, pp. 562–576, Apr. 2008.

[38] M. Oliveira, A. D. Sappa, and V. Santos, "A probabilistic approach for color correction in image mosaicking applications," *IEEE Trans. Image Process.*, vol. 24, no. 2, pp. 508–523, Feb. 2015.

[39] L. Yu, Y. Zhang, M. Sun, and X. Zhu, "Colour balancing of satellite imagery based on a colour reference library," *Int. J. Remote Sens.*, vol. 37, no. 24, pp. 5763–5785, Dec. 2016.

[40] Y. Boykov, O. Veksler, and R. Zabih, "Fast approximate energy minimization via graph cuts," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 23, no. 11, pp. 1222–1239, Nov. 2001.

[41] N. I. Fisher, T. Lewis, and B. J. Embleton, *Statistical Analysis of Spherical Data*, 1st ed. Cambridge, U.K.: Cambridge Univ. Press, 1993.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.

[42] X. Li, N. Hui, H. Shen, Y. Fu, and L. Zhang, "A robust mosaicking procedure for high spatial resolution remote sensing images," *J. Photogramm. Remote Sens.*, vol. 109, pp. 108–125, Nov. 2015.
[43] V. Rankov, R. J. Locke, R. J. Edens, P. R. Barber, and B. Vojnovic, "An algorithm for image stitching and blending," *Proc. SPIE*, vol. 5701, pp. 190–199, Mar. 2005.
[44] M. D. Patidar and A. Jain, "Automatic image mosaicing: An approach based on FFT," *Int. J. Sci. Eng. Technol.*, vol. 1, no. 1, pp. 1–4, Dec. 2011.
[45] M.-S. Su, W.-L. Hwang, and K.-Y. Cheng, "Variational calculus approach to multiresolution image mosaic," in *Proc. Int. Conf. Image Process.*, vol. 2, 2001, pp. 245–248.
[46] W. Zhao, "Flexible image blending for image mosaicing with reduced artifacts," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 20, no. 4, pp. 609–628, Jun. 2006.
[47] F. Zhang and F. Liu, "Parallax-tolerant image stitching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3262–3269.
[48] K. Lin, N. Jiang, L.-F. Cheong, M. Do, and J. Lu, "Seagull: Seam-guided local alignment for parallax-tolerant image stitching," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 370–385.
[49] J. Jia and C. K. Tang, "Image stitching using structure deformation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 4, pp. 617–631, Apr. 2008.
[50] G. Zhang, Y. He, W. Chen, J. Jia, and H. Bao, "Multi-viewpoint panorama construction with wide-baseline images," *IEEE Trans. Image Process.*, vol. 25, no. 7, pp. 3099–3111, Jul. 2016.
[51] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 4937–4946.
[52] M. Brown and D. G. Lowe, "Automatic panoramic image stitching using invariant features," *Int. J. Comput. Vis.*, vol. 74, no. 1, pp. 59–73, Aug. 2007.
[53] J. Zaragoza, T. Chin, Q. Tran, M. S. Brown, and D. Suter, "As-projective-as-possible image stitching with moving DLT," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 7, pp. 1285–1298, Jul. 2014.
[54] C.-C. Lin, S. U. Pankanti, K. N. Ramamurthy, and A. Y. Aravkin, "Adaptive as-natural-as-possible image stitching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1155–1163.
[55] Z. Zhang, "Parameter estimation techniques: A tutorial with application to conic fitting," *Image Vis. Comput.*, vol. 15, no. 1, pp. 59–76, 1997.
[56] K. Rohr *et al.*, "Point-based elastic registration of medical image data using approximating thin-plate splines," in *Proc. Int. Conf. Vis. Biomed. Comput.*, 1996, pp. 297–306.
[57] N. Arad and D. Reisfeld, "Image warping using few anchor points and radial functions," *Comput. Graph. Forum*, vol. 14, no. 1, pp. 35–46, Feb. 1995.
[58] J. Gao, Y. Li, T.-J. Chin, and M. S. Brown, "Seam-driven image stitching," in *Proc. Eurographics*, 2013, pp. 45–48.
[59] J. Li, Z. Wang, S. Lai, Y. Zhai, and M. Zhang, "Parallax-tolerant image stitching based on robust elastic warping," *IEEE Trans. Multimedia*, vol. 20, no. 7, pp. 1672–1687, Jul. 2018.
[60] T. Liao and N. Li, "Single-perspective warps in natural image stitching," *IEEE Trans. Image Process.*, vol. 29, pp. 724–735, 2020.
[61] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

Yan Mo received the B.Sc. degree from the Changchun University of Technology, Changchun, China, in 2015, and the M.S. degree from Xiangtan University, Xiangtan, China, in 2018. He is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

His research interests include image matching and image fusion.

Xudong Kang (Senior Member, IEEE) received the B.Sc. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interests include hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

Dr. Kang received the National Nature Science Award of China (Second Class and Rank as Third) and the Second Prize in the Student Paper Competition in IEEE Geoscience and Remote Sensing Society (IGARSS) 2014. He was selected as the Best Reviewer of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He also serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.

Puhong Duan (Student Member, IEEE) received the B.Sc. degree from Suzhou University, Suzhou, China, in 2014, and the M.S. degree from the Hefei University of Technology, Hefei, China, in 2017. He is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universität Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. His research interests include hyperspectral image classification, visualization, and image fusion.

Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or coauthored over 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is also a member of the Editorial Board of the *Information Fusion* and the *Sensing and Imaging*. He gained two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 09:51:39 UTC from IEEE Xplore. Restrictions apply.